

SOLUCIONARIO EXAMEN FINAL

ASIGNATURA:	QUÍMICA	PÁGINA:	1 de 2
FECHA:	30/12/2025		

Q1.- Sr, S, O

$$\text{Sr} = \frac{47,70}{88} = \frac{0,542}{0,542} = 1$$

$$\text{S} = \frac{17,45}{32} = \frac{0,545}{0,542} \approx 1$$

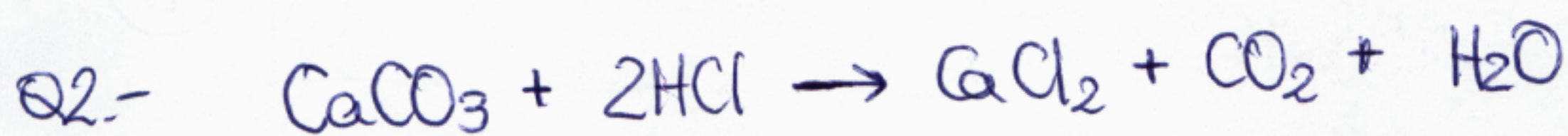
$$\text{O} = \frac{34,85}{16} = \frac{2,178}{0,542} = 4$$

F. Empírica: $(\text{Sr SO}_4)_x$

donde:

$$x = \frac{\bar{M} \cdot \text{FM}}{\bar{M} \cdot \text{FE}} = \frac{184}{184} = 1$$

Fórmula Molecular: $\boxed{\text{Sr SO}_4}$ (C)



$$V_{\text{CO}_2} = 15 \text{ L}$$

$$P_{\text{CO}_2} = 700 - 23,8 = 676,2 \text{ torr}$$

$$T_{\text{CO}_2} = 298 \text{ K}$$

$$P_{\text{CO}_2} \cdot V_{\text{CO}_2} = n_{\text{CO}_2} \cdot R \cdot T_{\text{CO}_2}$$

$$n_{\text{CO}_2} = \frac{P_{\text{CO}_2} \cdot V_{\text{CO}_2}}{R \cdot T_{\text{CO}_2}} = 0,545 \text{ mol CO}_2$$

$$0,545 \text{ mol CO}_2 \times \frac{1 \text{ mol CaCO}_3}{1 \text{ mol CO}_2} \times \frac{100 \text{ g CaCO}_3}{1 \text{ mol CaCO}_3} \times \frac{100 \text{ g caliza}}{85 \text{ g CaCO}_3} = \boxed{64,1 \text{ g caliza}} \text{ (A)}$$

Q3.- $M = \frac{f \cdot 100\%}{\bar{M}_{\text{H}_2\text{SO}_4}} = 16,71 \text{ M}$

$$C_1 V_1 = C_2 V_2$$

$$16,71 \cdot V_1 = 4,18 \cdot 500 \text{ ml} \Rightarrow \boxed{V_1 = 125 \text{ ml}} \text{ (B)}$$

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Q4.- $m_{\text{solute}} = 2 \text{ g sust.}$

$m_{\text{H}_2\text{O}} = 20 \text{ g H}_2\text{O} = 0,020 \text{ Kg H}_2\text{O}$

$T_{\text{eb}} = 100,6^\circ\text{C}$

$$\Delta T_{\text{eb}} = \frac{m_{\text{solute}} \cdot K_{\text{e}}}{\bar{M}_{\text{solute}} \cdot \text{Kg H}_2\text{O}} \Rightarrow (100,6 - 100)^\circ\text{C} = \frac{2 \text{ g} \cdot 0,52^\circ\text{C Kg/mol}}{\bar{M}_{\text{solute}} \cdot 0,020 \text{ Kg}}$$

$\bar{M}_{\text{solute}} = 86,67 \text{ g/mol} \approx \boxed{\bar{M}_{\text{solute}} = 87 \text{ g/mol}} \text{ (D)}$

Q5.- $\pi = C \cdot R \cdot T$

$\pi = 14,20 \text{ torr}$

$m_x = 0,194 \text{ g}$

$V_b = 82,4 \text{ ml} = 0,0824$

$R = 62,4 \frac{\text{torr} \cdot \text{L}}{\text{mol} \cdot \text{K}}$

$T = 25^\circ\text{C} = 298 \text{ K}$

$$\pi = \frac{m_x}{\bar{M}_x \cdot V} \cdot R \cdot T$$

$$\bar{M}_x = \frac{m_x \cdot R \cdot T}{\pi \cdot V}$$

$\bar{M}_x = 3083,1 \frac{\text{g}}{\text{mol}} \approx \boxed{3083 \text{ g/mol}} \text{ (A)}$